Cambridge Assessment International Education

Cambridge O Level

PHYSICS **5054/31**

Paper 3 Practical Test **October/November 2021**

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	‘List rule’ guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6	Calculation specific guidance Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	Guidance for chemical equations Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Examples of how to apply the list rule

State **three** reasons…. [3]

A

1	Correct	✓	**2**
2	Correct	✓	
3	Wrong	×	

B (4 responses)

1	Correct, Correct	✓, ✓	**3**
2	Correct	✓	
3	Wrong	ignore	

C (4 responses)

1	Correct	✓	**2**
2	Correct, Wrong	✓, ×	
3	Correct	ignore	

D (4 responses)

1	Correct	✓	**2**
2	Correct, CON (of 2.)	×, (discount 2)	
3	Correct	✓	

E (4 responses)

1	Correct	✓	**3**
2	Correct	✓	
3	Correct, Wrong	✓	

F (4 responses)

1	Correct	✓	**2**
2	Correct	✓	
3	Correct CON (of 3.)	× (discount 3)	

G (5 responses)

1	Correct	✓	**3**
2	Correct	✓	
3	Correct Correct CON (of 4.)	✓ ignore ignore	

H (4 responses)

1	Correct	✓	**2**
2	Correct	×	
3	CON (of 2.) Correct	(discount 2) ✓	

I (4 responses)

1	Correct	✓	**2**
2	Correct	×	
3	Correct CON (of 2.)	✓ (discount 2)	

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Question	Answer	Marks
1(a)	60°;	**1**
1(b)(i)	Correct construction lines on Fig. 1.3 **And** Angle b_x present and < 90° ;	**1**
1(b)(ii)	angle b_y < b_x **and** b_z < b_y and minimum accepted angle b_z = 50 ° ;	**1**
1(c)	the angle decreases (as ray moves from X to Y to Z) ;	**1**
1(d)	too much background light / ray / lamp too dim / ray spreads out (too much) / only possible to measure angles to within e.g. ± 2° / not possible to measure angles precisely / thickness of mirrors and reflecting surface at back / AVP ;	**1**

Question	Answer	Marks
2(a)	All components correctly positioned and workable circuit correct ;;; or any **two from**: switch in series with three cells ; **or** two lamps in series ; **or** LDR and fixed resistor in series ; **or** ammeter and voltmeter in correct positions ;	**3**
2(b)	0.16 A / amps **and** 0.7 V / volts (or SV / CV);	**1**
2(c)	(*the readings change when the light level changes so…*) keep still / don't block the light / don't cover the LDR / keep the lamps at the same distance from the LDR / keep the light level the same ;	**1**

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Question	Answer	Marks
3(a)(i)	average of at least two time measurements and in the range $t_1 = 3.0 \pm 1.0$ s ;	**1**
3(a)(ii)	$t_2 > t_1$ and $t_2 < 5$ s ;	**1**
3(b)	$V_1 = 50 \div t_1$ e.g. 19.4 cm / s $V_2 = 50 \div t_2$ e.g. 13.6 cm / s	**1**
3(c)	Two values produced and used, one derived from a further test run (of 100 cm) and one using the formula; Makes a correct comparison/conclusion whether the claim is true using those two values; i.e. evaluates V_3 using the formula e.g. 16.02 cm / s	**2**

Question	Answer	Marks
4(a)	**Two** from: there is a weight placed on each base bosses are attached as low as possible d is not too large / the spring is not too stretched ;;	**2**
4(b)	38 cm $\pm$ 4 cm ;	**1**
4(c)(i)	measure time for 10 oscillations and divide this by 10 **or** use a fiducial marker **or** count oscillations as the rod moves at its fastest in the same directions (through the resting point / centre of oscillation) all OWTTE ;	**1**
4(c)(ii)	(11.5 ÷ 10 =) 1.15 s $\pm$ 0.15 s from between 5 and 20 oscillations ;	**1**
4(d)	headings and units ; eight measurements with a range of at least 25 cm ; calculations of L^2 / 100 and T^2 – check the last completed row of the table ; consistent number of decimal places ;	**4**

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Question	Answer	Marks
4(e)	axes labelled with correct orientation (T^2 on y axis) ✓A suitable scale, plots occupying ⩾ ½ page in both directions ✓Q points plotted correctly check and circle/annotate two points ✓P best fit straight line ✓L	**4**
4(f)	**Either**: (Yes) the points at each end are below the best fit line and the middle points are above **or** the increase in T^2 gets smaller ; **or** (No) the points are all evenly spread about the best fit line **or** (No) the graph shows that T^2 increases by the same amount for equal increases in L^2 / 100 so the line should be straight (OWTTE) ;	**1**
4(g)	**Either** (Yes) reaction time is a smaller proportion of the total time when L is large **or** it's easier to judge the end of the swing when it is moving slower **or** points are closer to best fit line at large *L* ; **or** (No) reaction time is the same **or** points are closer to best fit line at smaller *L* ;	**1**

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